Finite-Time Generalized Advantage Estimation

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Abstract

Advantage estimation is a central component of actor-critic reinforcement learning algorithms, such as A2C or PPO. One of the most widely used methods is generalized advantage estimation (GAE). However, its adaptation to practical finite-time rollout implementations is not covered by the infinite-horizon theory commonly used to derive GAE. This work suggests to explicitly take into account the nature of finite-time rollouts to redefine the weighting of mixtures of k-step advantages. We analyze how standard truncated GAE changes the effective weighting of available estimators near rollout endpoints and propose *finite-time GAE (FT-GAE)*, which normalizes geometric weights over the available targets and retains a linear-time backward recursion. It is shown how our reweighting changes estimation error near rollout endpoints analytically under controlled bias-variance models and empirically in a controlled gridworld experiment. Extensive deep RL experiments across diverse benchmark families show substantial improvements when truncated GAE is replaced with FT-GAE. Controlled terminal-signal ablations provide further evidence that its advantage can be particularly pronounced when endpoint signals are informative.

1 Preliminaries

We consider a Markovian environment described by $(\mathcal{S}, \mathcal{A}, R, p)$ with initial distribution μ and time-stationary policy π. This induces a process $(S_t, A_t, R_t)_{0 \le t < T}$, where $T \in \mathbb{N} \cup \{\infty\}$. We allow either $T < \infty$ with $\gamma \in [0, 1]$, or $T = \infty$ with $\gamma \in [0, 1)$. For a policy π, define the value function $V_t^\pi(s) := \mathbb{E}^\pi[\sum_{i=t}^{T-1} \gamma^{i-t} R_i | S_t = s]$, state-action value function $Q_t^\pi(s, a) := \mathbb{E}^\pi[\sum_{i=t}^{T-1} \gamma^{i-t} R_i | S_t = s, A_t = a]$ and advantage function $\mathbb{A}_t^\pi(s, a) := Q_t^\pi(s, a) - V_t^\pi(s)$. For notational simplicity, and to stay close to PPO practice, we pretend the critic V is stationary. For a parameterized policy π_θ, the objective is $J(\theta) := V_0^{\pi_\theta}(\mu) := \int_{\mathcal{S}} V_0^{\pi_\theta}(s)\, d\mu(s)$. The policy-gradient theorem in the advantage form is $\nabla_\theta J(\theta) = \sum_{t=0}^{T-1} \gamma^t \mathbb{E}^{\pi_\theta}[\nabla_\theta \log \pi_\theta(A_t \mid S_t)\, \mathbb{A}_t^{\pi_\theta}(S_t, A_t)]$. Actor-critic methods approximate this expression by replacing the advantage with an estimator $\hat{\mathbb{A}}_t$ constructed from rollout data and a learned value function.

2 From k-step advantages to generalized advantage estimation

Let $V : \mathcal{S} \to \mathbb{R}$ be a fixed value function approximation. We first consider the infinite-horizon discounted setting with bounded rewards and bounded V. For a finite horizon, the same identities hold with time-dependent values V_t and terminal value $V_T = 0$; equivalently, time can be included in the state. The one-step temporal-difference error is

$$\delta_t := R_t + \gamma V(S_{t+1}) - V(S_t).$$

If $V = V^\pi$, then δ_t is conditionally unbiased for the advantage since

$$\begin{aligned}\mathbb{E}^\pi[\delta_t \mid S_t, A_t] &= \mathbb{E}^\pi[R_t + \gamma V^\pi(S_{t+1}) \,|\, S_t, A_t] - V^\pi(S_t) \\ &= Q^\pi(S_t, A_t) - V^\pi(S_t) \\ &= \mathbb{A}^\pi(S_t, A_t)\end{aligned}$$

due to Bellmann equation. With an approximate critic, this estimator is biased through the bootstrap term $V(S_{t+1})$.

More generally, for $k \geq 0$, define the k-step return and the corresponding advantage estimator by

$$\hat{G}_t^{(k)} := \sum_{\ell=0}^{k} \gamma^\ell R_{t+\ell} + \gamma^{k+1} V(S_{t+k+1}), \qquad \hat{\mathbb{A}}_t^{(k)} := \hat{G}_t^{(k)} - V(S_t).$$

Thus, $k = 0$ corresponds to one sampled reward and $\hat{\mathbb{A}}_t^{(0)} = \delta_t$.

When $V = V^\pi$, each $\hat{\mathbb{A}}_t^{(k)}$ again is conditionally unbiased for $\mathbb{A}^\pi(S_t, A_t)$ due to k-step bellman equation. The parameter k controls the standard bias-variance trade-off: small k relies more on bootstrapping, typically lowering variance but increasing critic-induced bias, whereas large k uses longer sampled returns, reducing bootstrap bias but increasing variance; as $k \to \infty$, one recovers the Monte Carlo advantage whenever the discounted return is well defined.

In addition, the estimator admits a TD-error representation:

$$\begin{aligned}&\sum_{\ell=0}^{k} \gamma^\ell \delta_{t+\ell} = \sum_{\ell=0}^{k} \gamma^\ell R_{t+\ell} + \sum_{\ell=0}^{k} \gamma^{\ell+1} V(S_{t+\ell+1}) - \sum_{\ell=0}^{k} \gamma^\ell V(S_{t+\ell}) \\ &= \sum_{\ell=0}^{k} \gamma^\ell R_{t+\ell} + \sum_{\ell=1}^{k+1} \gamma^\ell V(S_{t+\ell}) - \sum_{\ell=0}^{k} \gamma^\ell V(S_{t+\ell}) \\ &= \sum_{\ell=0}^{k} \gamma^\ell R_{t+\ell} + \left[\sum_{\ell=1}^{k} \gamma^\ell V(S_{t+\ell}) + \gamma^{k+1} V(S_{t+k+1})\right] - \left[V(S_t) + \sum_{\ell=1}^{k} \gamma^\ell V(S_{t+\ell})\right] \\ &= \sum_{\ell=0}^{k} \gamma^\ell R_{t+\ell} + \gamma^{k+1} V(S_{t+k+1}) - V(S_t) \\ &= \hat{\mathbb{A}}_t^{(k)}.\end{aligned}$$

Remark 2.1 (Indexing convention). The original GAE paper [2] defines the k-step advantage with bootstrapping at time $t+k$. In contrast, our definition in Section 1 bootstraps at time $t+k+1$. Equivalently, our k-step estimator corresponds to the $(k+1)$-step estimator in the indexing used in GAE. This is purely a notational shift chosen so that geometric mixtures take the form $\sum_{k\geq 0} \lambda^k \hat{\mathbb{A}}_t^{(k)}$.

For $\lambda \in (0,1)$, the infinite-horizon λ-return and generalized advantage estimators [2] are the mixtures

$$\hat{G}_t^{\infty,\lambda} := (1-\lambda)\sum_{k=0}^{\infty} \lambda^k \hat{G}_t^{(k)}$$

and

$$\hat{\mathbb{A}}_t^{\infty,\lambda} := (1-\lambda)\sum_{k=0}^{\infty} \lambda^k \hat{\mathbb{A}}_t^{(k)}.$$

The weight normalization is chosen such that $(1-\lambda)\sum_{k=0}^{\infty}\lambda^k = 1$. This estimator admits a TD-error representation: Using the identity $\hat{\mathbb{A}}_t^{(k)} = \sum_{\ell=0}^{k}\gamma^\ell \delta_{t+\ell}$ yields

$$\hat{\mathbb{A}}_t^{\infty,\lambda} = (1-\lambda)\sum_{k=0}^{\infty}\lambda^k \sum_{\ell=0}^{k}\gamma^\ell\delta_{t+\ell} = (1-\lambda)\sum_{\ell=0}^{\infty}\gamma^\ell\delta_{t+\ell}\sum_{k=\ell}^{\infty}\lambda^k = \sum_{\ell=0}^{\infty}(\gamma\lambda)^\ell\delta_{t+\ell}.$$

Thus, future TD errors are weighted geometrically by the product $\gamma\lambda$. Smaller λ yields more bootstrapping and typically lower variance but greater sensitivity to critic bias, and vice versa.

The preceding definitions are infinite-horizon objects: they require $\delta_{t+\ell}$ to be defined for all future times. For finite rollouts, PPO instead uses truncated GAE [3]. If the rollout ends at τ, then

$$\hat{\mathbb{A}}_t^{\mathrm{trunc},\lambda} := \sum_{\ell=0}^{\tau-t-1}(\gamma\lambda)^\ell\,\delta_{t+\ell}, \qquad t \leq \tau-1.$$

This estimator can be computed backward as

$$\hat{\mathbb{A}}_t^{\mathrm{trunc},\lambda} = \sum_{\ell=0}^{\tau-t-1}(\gamma\lambda)^\ell\delta_{t+\ell} = \delta_t + \gamma\lambda\sum_{\ell=0}^{\tau-t-2}(\gamma\lambda)^\ell\delta_{t+1+\ell} = \delta_t + \gamma\lambda\hat{\mathbb{A}}_{t+1}^{\mathrm{trunc},\lambda}, \qquad \hat{\mathbb{A}}_\tau^{\mathrm{trunc},\lambda} := 0.$$

However, the original interpretation of GAE as a geometric mixture of k-step estimators with weights $(1-\lambda)\lambda^k$ no longer holds unchanged. Instead, all geometric mass corresponding to unavailable k-step estimators beyond the rollout end is assigned to the longest available estimator.

Proposition 2.2 (Tail-mass collapse of truncated GAE)**.** *For $t \leq \tau-1$, truncated GAE satisfies*

$$\hat{\mathbb{A}}_t^{\mathrm{trunc},\lambda} = \sum_{k=0}^{\tau-t-2}\underbrace{(1-\lambda)\lambda^k}_{\textit{GAE weights}}\hat{\mathbb{A}}_t^{(k)} + \underbrace{\lambda^{\tau-t-1}}_{\textit{collapsed tail-mass}}\hat{\mathbb{A}}_t^{(\tau-t-1)}. \tag{1}$$

Proof. Fix $t \in \{0, \dots, \tau-1\}$. Let $\tilde{\mathbb{A}}_t := (1-\lambda) \sum_{k=0}^{\tau-t-2} \lambda^k \hat{\mathbb{A}}_t^{(k)} + \lambda^{\tau-t-1} \hat{\mathbb{A}}_t^{(\tau-t-1)}$. We prove that the identity for $\tilde{\mathbb{A}}_t$ exactly yields truncated GAE. Since $\hat{\mathbb{A}}_t^{(k)} = \sum_{\ell=0}^{k} \gamma^\ell \delta_{t+\ell}$, we obtain

$$\tilde{\mathbb{A}}_t = (1-\lambda) \sum_{k=0}^{\tau-t-2} \lambda^k \sum_{\ell=0}^{k} \gamma^\ell \delta_{t+\ell} \; + \; \lambda^{\tau-t-1} \sum_{\ell=0}^{\tau-t-1} \gamma^\ell \delta_{t+\ell}.$$

For the first term we swap the order of summation and use standard formulas for

$$\begin{aligned}
(1-\lambda) \sum_{k=0}^{\tau-t-2} \lambda^k \sum_{\ell=0}^{k} \gamma^\ell \delta_{t+\ell} &= (1-\lambda) \sum_{k=0}^{\tau-t-2} \lambda^k \sum_{\ell=0}^{\tau-t-2} \mathbf{1}_{\{\ell \le k\}} \gamma^\ell \delta_{t+\ell} \\
&= (1-\lambda) \sum_{\ell=0}^{\tau-t-2} \gamma^\ell \delta_{t+\ell} \sum_{k=\ell}^{\tau-t-2} \lambda^k \\
&= (1-\lambda) \sum_{\ell=0}^{\tau-t-2} \gamma^\ell \delta_{t+\ell} \, \lambda^\ell \sum_{j=0}^{\tau-t-\ell-2} \lambda^j \\
&= \sum_{\ell=0}^{\tau-t-2} \gamma^\ell \delta_{t+\ell} \, \lambda^\ell \big(1 - \lambda^{\tau-t-\ell-1}\big) \\
&= \sum_{\ell=0}^{\tau-t-2} (\gamma\lambda)^\ell \delta_{t+\ell} \; - \; \lambda^{\tau-t-1} \sum_{\ell=0}^{\tau-t-2} \gamma^\ell \delta_{t+\ell}.
\end{aligned}$$

Plugging this back into the formula for $\tilde{\mathbb{A}}_t$, yields

$$\begin{aligned}
\tilde{\mathbb{A}}_t &= \sum_{\ell=0}^{\tau-t-2} (\gamma\lambda)^\ell \delta_{t+\ell} - \lambda^{\tau-t-1} \sum_{\ell=0}^{\tau-t-2} \gamma^\ell \delta_{t+\ell} + \lambda^{\tau-t-1} \sum_{\ell=0}^{\tau-t-1} \gamma^\ell \delta_{t+\ell} \\
&= \sum_{\ell=0}^{\tau-t-2} (\gamma\lambda)^\ell \delta_{t+\ell} + \lambda^{\tau-t-1} \gamma^{\tau-t-1} \delta_{t+\tau-t-1} = \sum_{\ell=0}^{\tau-t-1} (\gamma\lambda)^\ell \delta_{t+\ell}.
\end{aligned}$$

This is exactly truncated GAE. □

We call the effect of Proposition 2.2 *tail-mass collapse*; see the blue bars in Figure 1. In a finite rollout, estimators beyond τ are unavailable, so truncated GAE places this entire tail mass on the longest available estimator $\hat{\mathbb{A}}_t^{(\tau-t-1)}$. Consequently, advantage estimators close to the rollout end become substantially more Monte-Carlo-like than the nominal value of λ suggests. Hence, the amount of collapsed mass depends on the remaining rollout length $\tau - t$, so the effective bias-variance trade-off near the endpoint varies locally through the distance to τ, not only by the chosen value of λ. In actor-critic implementations, the endpoint τ is not only determined by the finite rollout buffer, and may correspond either to a true termination or to an artificial rollout endpoint.

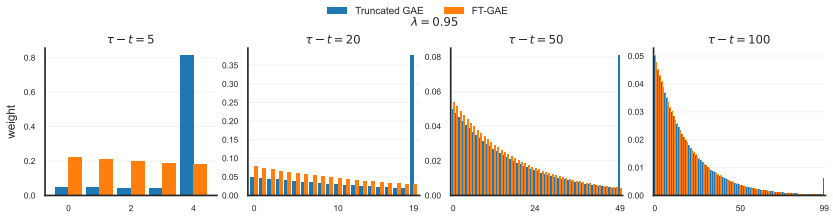


Figure 1: **GAE tail-mass collapse.** Comparison of the weights assigned to the k-step advantage estimators $\hat{\mathbb{A}}_t^{(k)}$ by truncated GAE (blue) and our FT-GAE introduced in Definition 3.1 (orange) for $\lambda = 0.95$. Each panel corresponds to a different remaining horizon $\tau - t$ and shows how, for truncated GAE, the tail mass collapses onto the last available estimator $\hat{\mathbb{A}}_t^{(\tau-t-1)}$, while the finite-time variant redistributes this mass over the observable k-step estimators via finite-horizon renormalization.

3 Mitigating tail-mass collapse: Finite-time GAE (FT-GAE)

Why can tail-mass collapse be problematic in RL? Solving (discrete) optimal control problems is closely related to dynamic programming and as such should profit from bootstrapping future information. This is inherent in value-based methods such as Q-learning but less so in policy gradient methods (high λ values). As an example it is surprising how little PPO training of Lunar Lander profits from the strong terminal signal at crashing/landing. A reason we identified is that tail-mass collapse leads to excessive Monte Carlo close to termination instead of bootstrapping already learned information. As termination can mean crashing (-100) or landing (+100), this significantly increases variance of the corresponding truncated GAE estimators. More generally, tail-mass collapse can be costly when a terminal reward is salient relative to within-episode rewards and its realized outcome varies across trajectories. If nearby value estimates already capture its expected effect, shorter bootstrapped targets retain the expected signal while decreasing variance.

Require: Policy π_θ, value estimate V, discount γ, GAE parameter λ

1: Generate rollout $\{(s_t, a_t, r_t, v_t)\}_{t=0}^{\tau-1}$ with $v_t \leftarrow V(s_t)$ for $t = 0, \dots, \tau$

2: $\hat{\mathbb{A}}_\tau^{\mathrm{FT}} \leftarrow 0$

3: **for** $t = \tau - 1, \dots, 0$ **do**

4: $\quad \delta_t \leftarrow r_t + \gamma v_{t+1} - v_t$

5: $\quad \hat{\mathbb{A}}_t^{\mathrm{FT}} \leftarrow \delta_t + \gamma\lambda \dfrac{1-\lambda^{\tau-t-1}}{1-\lambda^{\tau-t}} \hat{\mathbb{A}}_{t+1}^{\mathrm{FT}}$

6: **end for**

7: **return** $\{\hat{\mathbb{A}}_t^{\mathrm{FT}}\}_{t=0}^{\tau-1}$

Algorithm 1: Finite-time GAE (ours)

We propose a simple modification, *finite-time GAE*, that removes tail-mass collapse by dividing the geometric weights of the available k-step estimators at each rollout position by their finite sum.

Definition 3.1 (Finite-time GAE)**.** For $\lambda \neq 1$, we define the finite-time general-

ized advantage estimator (FT-GAE) by

$$\hat{\mathbb{A}}_t^{\mathrm{FT},\lambda} := \frac{1-\lambda}{1-\lambda^{\tau-t}} \sum_{k=0}^{\tau-t-1} \lambda^k \, \hat{\mathbb{A}}_t^{(k)}, \qquad t < \tau. \tag{2}$$

For $\lambda = 1$, this mixture is understood as its continuous extension, yielding the uniform average.

Finite-time GAE preserves the geometric preference for shorter look-aheads, but normalizes this preference over the finite set $k \in \{0, \ldots, \tau - t - 1\}$. Thus, both truncated GAE and FT-GAE are position-dependent on finite rollouts, but in different ways: truncated GAE collapses the unavailable tail mass onto the longest available estimator, whereas FT-GAE redistributes this mass over all available estimators according to the normalized finite geometric law. The orange bars in Figure 1 show these weights. Consequently, FT-GAE differs appreciably from truncated GAE only close to rollout endpoints, where the missing tail mass is largest.

Finite-time normalization also changes the limiting behavior as $\lambda \uparrow 1$. For truncated GAE, the limiting mixture puts all mass on the longest available estimator $\hat{\mathbb{A}}_t^{(\tau-t-1)}$. In contrast, FT-GAE remains a normalized finite mixture: at $\lambda = 1$, it is the uniform average over all available k-step estimators, not the Monte Carlo estimator. Although FT-GAE can be extended to $\lambda > 1$, assigning increasing weight to larger k and recovering a Monte Carlo-type limit as $\lambda \to \infty$, we focus on the practically relevant regime $\lambda \leq 1$. Appendix Figure 2 compares the resulting weighting schemes.

FT-GAE also admits a linear-time backward pass, formalized in the following proposition:

Proposition 3.2 (FT-GAE recursion)**.** *For $\lambda \neq 1$, FT-GAE admits the TD-sum representation*

$$\hat{\mathbb{A}}_t^{\mathrm{FT},\lambda} = \sum_{\ell=0}^{\tau-t-1} (\gamma\lambda)^\ell \, \frac{1-\lambda^{\tau-t-\ell}}{1-\lambda^{\tau-t}} \, \delta_{t+\ell}$$

for $t < \tau$, and satisfies the backward recursion

$$\hat{\mathbb{A}}_t^{\mathrm{FT},\lambda} = \delta_t + \gamma\lambda \, \frac{1-\lambda^{\tau-t-1}}{1-\lambda^{\tau-t}} \, \hat{\mathbb{A}}_{t+1}^{\mathrm{FT},\lambda}, \qquad \hat{\mathbb{A}}_\tau^{\mathrm{FT},\lambda} := 0.$$

Proof. Fix $t \in \{0, \dots, \tau - 1\}$.

$$\begin{aligned}
\hat{\mathbb{A}}_t^{\mathrm{FT},\lambda} &= \frac{1-\lambda}{1-\lambda^{\tau-t}} \sum_{k=0}^{\tau-t-1} \lambda^k \hat{\mathbb{A}}_t^{(k)} = \frac{1-\lambda}{1-\lambda^{\tau-t}} \sum_{k=0}^{\tau-t-1} \lambda^k \sum_{\ell=0}^{\tau-t-1} \mathbf{1}_{\{\ell \le k\}} \gamma^\ell \delta_{t+\ell} \\
&= \frac{1-\lambda}{1-\lambda^{\tau-t}} \sum_{\ell=0}^{\tau-t-1} \gamma^\ell \delta_{t+\ell} \sum_{k=\ell}^{\tau-t-1} \lambda^k = \frac{1-\lambda}{1-\lambda^{\tau-t}} \sum_{\ell=0}^{\tau-t-1} \gamma^\ell \delta_{t+\ell}\, \lambda^\ell \sum_{k=0}^{\tau-t-\ell-1} \lambda^k \\
&= \frac{1-\lambda}{1-\lambda^{\tau-t}} \sum_{\ell=0}^{\tau-t-1} \gamma^\ell \delta_{t+\ell}\, \lambda^\ell \frac{1-\lambda^{\tau-t-\ell}}{1-\lambda} = \frac{1-\lambda}{1-\lambda^{\tau-t}} \sum_{\ell=0}^{\tau-t-1} \gamma^\ell \delta_{t+\ell} \frac{\lambda^\ell - \lambda^{\tau-t}}{1-\lambda} \\
&= \sum_{\ell=0}^{\tau-t-1} \frac{\lambda^\ell - \lambda^{\tau-t}}{1-\lambda^{\tau-t}} \gamma^\ell \delta_{t+\ell} = \sum_{\ell=0}^{\tau-t-1} \frac{1-\lambda^{\tau-t-\ell}}{1-\lambda^{\tau-t}} \delta_{t+\ell} (\gamma\lambda)^\ell.
\end{aligned}$$

Thus, we can use above equation to obtain the recursive formula via induction in t. The base case follows easily with $\hat{\mathbb{A}}_\tau^{\mathrm{FT},\lambda} = 0$ as for $t = \tau - 1$, we have $\hat{\mathbb{A}}_{\tau-1}^{\mathrm{FT},\lambda} = \delta_{\tau-1}$. For general $t \in \{0, \dots, \tau - 2\}$, above formula yields

$$\begin{aligned}
\hat{\mathbb{A}}_t^{\mathrm{FT},\lambda} &= \delta_t + \sum_{\ell=1}^{\tau-t-1} \frac{\lambda^\ell - \lambda^{\tau-t}}{1-\lambda^{\tau-t}} \gamma^\ell \delta_{t+\ell} \\
&= \delta_t + \sum_{\ell=0}^{\tau-t-2} \frac{\lambda^{\ell+1} - \lambda^{\tau-t}}{1-\lambda^{\tau-t}} \gamma^{\ell+1} \delta_{t+1+\ell} \\
&= \delta_t + \frac{1-\lambda^{\tau-t-1}}{1-\lambda^{\tau-t}} \lambda\gamma\, \hat{\mathbb{A}}_{t+1}^{\mathrm{FT},\lambda}.
\end{aligned}$$

□

Relative to truncated GAE, only the coefficient of the recursive term changes in Algorithm 1: the fixed factor $\gamma\lambda$ becomes the horizon-dependent factor in Proposition 3.2. Thus, existing PPO implementations such as Stable-Baselines3 [1] require only a local modification of the GAE recursion. Implementation details can be found in Appendix **??**.

The case $\lambda = 1$ and weight-profiles. The finite-time normalization changes the limiting estimator at $\lambda = 1$. For truncated GAE, Proposition 2.2 gives

$$\hat{\mathbb{A}}_t^{\mathrm{trunc},1} = \hat{\mathbb{A}}_t^{(\tau-t-1)},$$

since all weights except the collapsed endpoint weight vanish. Thus truncated GAE becomes the longest available estimator.

In contrast, FT-GAE is defined at $\lambda = 1$ by the continuous extension of its normalized finite geometric weights:

$$\hat{\mathbb{A}}_t^{\mathrm{FT},1} = \frac{1}{\tau - t} \sum_{k=0}^{\tau-t-1} \hat{\mathbb{A}}_t^{(k)}.$$

Hence FT-GAE at $\lambda = 1$ is not the longest available estimator, but the uniform average over all observable k-step estimators. In that case we can derive the analogue to Proposition 3.2.

Proposition 3.3. *For $t < \tau$, the uniform FT-GAE estimator $\hat{\mathbb{A}}_t^{\mathrm{FT},1}$ admits the TD-sum representation*

$$\hat{\mathbb{A}}_t^{\mathrm{FT},1} = \sum_{\ell=0}^{\tau-t-1} \gamma^\ell \frac{\tau-t-\ell}{\tau-t} \delta_{t+\ell}.$$

Moreover, with $\hat{\mathbb{A}}_\tau^{\mathrm{FT},1} := 0$, it satisfies the backward recursion

$$\hat{\mathbb{A}}_t^{\mathrm{FT},1} = \delta_t + \gamma \frac{\tau-t-1}{\tau-t} \hat{\mathbb{A}}_{t+1}^{\mathrm{FT},1}.$$

Proof. Using

$$\hat{\mathbb{A}}_t^{(k)} = \sum_{\ell=0}^{k} \gamma^\ell \delta_{t+\ell},$$

we obtain

$$\hat{\mathbb{A}}_t^{\mathrm{FT},1} = \frac{1}{\tau-t} \sum_{k=0}^{\tau-t-1} \hat{\mathbb{A}}_t^{(k)} = \frac{1}{\tau-t} \sum_{k=0}^{\tau-t-1} \sum_{\ell=0}^{k} \gamma^\ell \delta_{t+\ell}.$$

Exchanging the order of summation gives

$$\begin{aligned}
\hat{\mathbb{A}}_t^{\mathrm{FT},1} &= \frac{1}{\tau-t} \sum_{\ell=0}^{\tau-t-1} \gamma^\ell \delta_{t+\ell} \sum_{k=\ell}^{\tau-t-1} 1 \\
&= \sum_{\ell=0}^{\tau-t-1} \gamma^\ell \frac{\tau-t-\ell}{\tau-t} \delta_{t+\ell}.
\end{aligned}$$

This proves the TD-sum representation.

For the recursion, separate the first TD error:

$$\hat{\mathbb{A}}_t^{\mathrm{FT},1} = \delta_t + \sum_{\ell=1}^{\tau-t-1} \gamma^\ell \frac{\tau-t-\ell}{\tau-t} \delta_{t+\ell}.$$

With the change of index $r = \ell - 1$, this becomes

$$\begin{aligned}
\hat{\mathbb{A}}_t^{\mathrm{FT},1} &= \delta_t + \gamma \sum_{r=0}^{\tau-t-2} \gamma^r \frac{\tau-t-1-r}{\tau-t} \delta_{t+1+r} \\
&= \delta_t + \gamma \frac{\tau-t-1}{\tau-t} \sum_{r=0}^{\tau-t-2} \gamma^r \frac{\tau-t-1-r}{\tau-t-1} \delta_{t+1+r}. \\
&= \delta_t + \gamma \frac{\tau-t-1}{\tau-t} \hat{\mathbb{A}}_{t+1}^{\mathrm{FT},1}.
\end{aligned}$$

□

We now use Propositions 2.2 to directly relate the estimators.

Corollary 3.4 (Truncated GAE as a mixture of FT-GAE and the longest target)**.** *Fix $t \leq \tau - 1$. Then*

$$\hat{\mathbb{A}}_t^{\mathrm{trunc},\lambda} = (1 - \lambda^{\tau-t})\hat{\mathbb{A}}_t^{\mathrm{FT},\lambda} + \lambda^{\tau-t}\hat{\mathbb{A}}_t^{(\tau-t-1)}. \tag{3}$$

For $\lambda \in [0, 1]$, this is a convex combination of FT-GAE and the longest observable k-step estimator.

Proof. By definition of FT-GAE,

$$(1 - \lambda^{\tau-t})\hat{\mathbb{A}}_t^{\mathrm{FT},\lambda} = (1 - \lambda) \sum_{k=0}^{\tau-t-1} \lambda^k \hat{\mathbb{A}}_t^{(k)}.$$

Therefore,

$$\begin{aligned}
&(1 - \lambda^{\tau-t})\hat{\mathbb{A}}_t^{\mathrm{FT},\lambda} + \lambda^{\tau-t}\hat{\mathbb{A}}_t^{(\tau-t-1)} \\
&\quad = (1 - \lambda) \sum_{k=0}^{\tau-t-2} \lambda^k \hat{\mathbb{A}}_t^{(k)} + (1 - \lambda)\lambda^{\tau-t-1}\hat{\mathbb{A}}_t^{(\tau-t-1)} + \lambda^{\tau-t}\hat{\mathbb{A}}_t^{(\tau-t-1)} \\
&\quad = (1 - \lambda) \sum_{k=0}^{\tau-t-2} \lambda^k \hat{\mathbb{A}}_t^{(k)} + \lambda^{m-1}\hat{\mathbb{A}}_t^{(\tau-t-1)} \\
&\quad = \hat{\mathbb{A}}_t^{\mathrm{trunc},\lambda},
\end{aligned}$$

where the last equality follows from Proposition 2.2. □

Equation (3) gives another view of tail-mass collapse. Truncated GAE equals the normalized finite geometric mixture used by FT-GAE, plus additional mass $\lambda^{\tau-t}$ placed on the longest observable k-step estimator. Thus, for large λ or short rollout suffixes $\tau - t$, truncated GAE becomes substantially more Monte-Carlo-like near the rollout endpoints. FT-GAE removes exactly this additional mass while preserving the relative geometric weighting over the observable horizons. Figure 2 visualizes the comparison of the weighting schemes for truncated GAE and our FT-GAE for different values of λ.

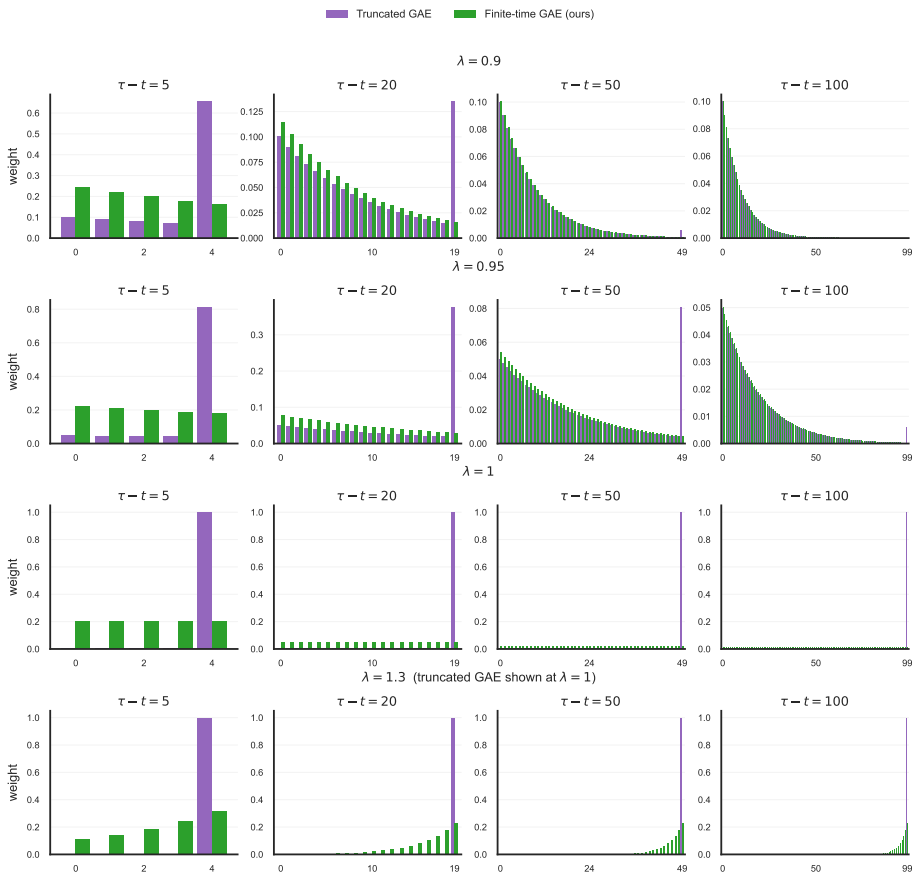


Figure 2: Weight profiles for truncated GAE and FT-GAE across different values of λ (rows) and remaining horizons $\tau - t$ (columns). For $\lambda \leq 1$, truncated GAE exhibits the tail-mass collapse described in Proposition 2.2: lowering λ reduces the collapsed endpoint mass, but changes the estimator globally and does not remove the underlying effect. FT-GAE instead forms a normalized finite mixture over the available k-step estimators. At $\lambda = 1$, this mixture is uniform. For $\lambda > 1$, it remains well-defined, with nonnegative weights that sum to one and increase with k, thereby emphasizing longer lookahead estimators. Truncated GAE has no analogous normalized weight profile for $\lambda > 1$. In those rows, we therefore show its $\lambda = 1$ profile as a reference.

References

[1] Antonin Raffin, Ashley Hill, Adam Gleave, Anssi Kanervisto, Maximilian Ernestus, and Noah Dormann. Stable-baselines3: reliable reinforcement learning implementations. *J. Mach. Learn. Res.*, 22(1), January 2021.

[2] John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation, 2018.

[3] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.